

Q1 - 24 June - Shift 1

The domain of the function

$$f(x) = \frac{\cos^{-1}\left(\frac{x^2 - 5x + 6}{x^2 - 9}\right)}{\log_e(x^2 - 3x + 2)} \text{ is}$$

(A) $(-\infty, 1) \cup (2, \infty)$

(B) $(2, \infty)$

(C) $\left[-\frac{1}{2}, 1\right) \cup (2, \infty)$

(D) $\left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{\frac{3 + \sqrt{5}}{2}, \frac{3 - \sqrt{5}}{2}\right\}$

Space for your notes:

Q2 - 24 June - Shift 1

The number of one-one function $f : \{a, b, c, d\} \rightarrow$

Space for your notes:

 $\{0, 1, 2, \dots, 10\}$ such that $2f(a) - f(b) + 3f(c) + f(d) = 0$ is _____.

Q3 - 25 June - Shift 1

Questions

MathonGo

Let $f : \mathbb{N} \rightarrow \mathbb{R}$ be a function such that $f(x+y) = 2f(x)f(y)$ for natural numbers x and y . If $f(1) = 2$, then the value of α for which

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$$\sum_{k=1}^{10} f(\alpha + k) = \frac{512}{3}(2^{20} - 1)$$

holds, is

- (A) 2 (B) 3
(C) 4 (D) 6

Q4 - 25 June - Shift 1

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

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$$f(x) = \left(2 \left(1 - \frac{x^{25}}{2} \right) (2 + x^{25}) \right)^{\frac{1}{50}}. \text{ If the function}$$

$g(x) = f(f(f(x))) + f(f(x))$, the the greatest integer less than or equal to $g(1)$ is _____

Q5 - 26 June - Shift 1

Let $f(x) = \frac{x-1}{x+1}$, $x \in \mathbb{R} - \{0, -1, 1\}$. If $f^{n+1}(x) = f(f^n(x))$

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for all $n \in \mathbb{N}$, then $f^6(6) + f^7(7)$ is equal to:

- (A) $\frac{7}{6}$ (B) $-\frac{3}{2}$ (C) $\frac{7}{12}$ (D) $-\frac{11}{12}$

Q6 - 26 June - Shift 2

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Questions

MathonGo

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined as $f(x) = x-1$ and

$g: \mathbb{R} - \{1, -1\} \rightarrow \mathbb{R}$ be defined as $g(x) = \frac{x^2}{x^2 - 1}$.

Then the function fg is :

- (A) one-one but not onto function
- (B) onto but not one-one function
- (C) both one-one and onto function
- (D) neither one-one nor onto function

Space for your notes:

Q7 - 27 June - Shift 1

Let $f: R \rightarrow R$ be a function defined $f(x) = \frac{2e^{2x}}{e^{2x} + e}$.

Then $f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \dots + f\left(\frac{99}{100}\right)$ is equal to _____.

Space for your notes:

Q8 - 27 June - Shift 2

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Define

$f: S \rightarrow S$ as $f(n) = \begin{cases} 2n, & \text{if } n = 1, 2, 3, 4, 5 \\ 2n-11 & \text{if } n = 6, 7, 8, 9, 10 \end{cases}$.

Let $g: S \rightarrow S$ be a function such that

$fg(n) = \begin{cases} n+1, & \text{if } n \text{ is odd} \\ n-1, & \text{if } n \text{ is even} \end{cases}$, then

$g(10) ((g(1) + g(2) + g(3) + g(4) + g(5)))$ is equal to:

Space for your notes:

Q9 - 28 June - Shift 1

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Let a function $f : \mathbb{N} \rightarrow \mathbb{N}$ be defined by

Space for your notes:

$$f(n) = \begin{cases} 2n, & n = 2, 4, 6, 8, \dots \\ n-1, & n = 3, 7, 11, 15, \dots \\ \frac{n+1}{2}, & n = 1, 5, 9, 13, \dots \end{cases}$$

then, f is

- (A) one-one but not onto
- (B) onto but not one-one
- (C) neither one-one nor onto
- (D) one-one and onto

Q10 - 28 June - Shift 2

The probability that a randomly chosen one-one function from the set $\{a, b, c, d\}$ to the set $\{1, 2, 3, 4, 5\}$ satisfies $f(a) + 2f(b) - f(c) = f(d)$ is :

Space for your notes:

- (A) $\frac{1}{24}$
- (B) $\frac{1}{40}$
- (C) $\frac{1}{30}$
- (D) $\frac{1}{20}$

Q11 - 28 June - Shift 2

Let $S = \{1, 2, 3, 4\}$. Then the number of elements in the set $\{f : S \times S \rightarrow S : f \text{ is onto and } f(a, b) = f(b, a) \geq a \forall (a, b) \in S \times S\}$ is

Space for your notes:

Q12 - 29 June - Shift 1

The domain of the function $\cos^{-1}\left(\frac{2\sin^{-1}\left(\frac{1}{4x^2-1}\right)}{\pi}\right)$

is :

- (A) $R - \left\{-\frac{1}{2}, \frac{1}{2}\right\}$
- (B) $(-\infty, -1] \cup [1, \infty) \cup \{0\}$
- (C) $\left(-\infty, \frac{-1}{2}\right) \cup \left(\frac{1}{2}, \infty\right) \cup \{0\}$
- (D) $\left(-\infty, \frac{-1}{\sqrt{2}}\right] \cup \left[\frac{1}{\sqrt{2}}, \infty\right) \cup \{0\}$

Space for your notes:

Q13 - 29 June - Shift 1

Let $c, k \in \mathbb{R}$. If $f(x) = (c+1)x^2 + (1-c^2)x + 2k$

and $f(x+y) = f(x) + f(y) - xy$, for all $x, y \in \mathbb{R}$, then

the value of $|2(f(1) + f(2) + f(3) + \dots + f(20))|$ is equal to _____.

Space for your notes:

Q14 - 29 June - Shift 2

Let $f(x)$ and $g(x)$ be two real polynomials of degree

2 and 1 respectively. If $f(g(x)) = 8x^2 - 2x$, and

$g(f(x)) = 4x^2 + 6x + 1$, then the value of $f(2) + g(2)$ is _____.

Space for your notes:

Answer Key

Q1 (D)

Q2 (31)

Q3 (C)

Q4 (2)

Q5 (B)

Q6 (D)

Q7 (99)

Q8 (190)

Q9 (D)

Q10 (D)

Q11 (37)

Q12 (D)

Q13 (3395)

Q14 (18)

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Q1 (D)

$$-1 \leq \frac{x^2 - 5x + 6}{x^2 - 9} \leq 1$$

$$\frac{x^2 - 5x + 6}{x^2 - 9} - 1 \leq 0$$

$$\frac{1}{x+3} \geq 0$$

$$x \in (-3, \infty) \dots\dots(1)$$

$$\frac{x^2 - 5x + 6}{x^2 - 9} + 1 \geq 0$$

$$\frac{2x+1}{x+3} \geq 0$$

$$x \in (-\infty, -3) \cup \left[-\frac{1}{2}, \infty\right) \dots\dots(2)$$

after taking intersection

$$x \in \left[-\frac{1}{2}, \infty\right)$$

$$x^2 - 3x + 2 > 0$$

$$x \in (-\infty, 1) \cup (2, \infty)$$

$$x^2 - 3x + 2 \neq 1$$

$$x \neq \frac{3 \pm \sqrt{5}}{2}$$

after taking intersection of each solution

$$\left[-\frac{1}{2}, 1\right) \cup (2, \infty) - \left\{\frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}\right\}$$

Q2 (31)

$$2f(a) + 3f(c) = f(d) - f(b)$$

Using fundamental principle of counting

Number of one-one function is 31

Q3 (C)

Hints and Solutions

MathonGo

$$f : \mathbb{N} \rightarrow \mathbb{R}, f(x+y) = 2f(x)f(y) \dots (1)$$

$$f(1) = 2,$$

$$\sum_{k=1}^{10} f(\alpha+k) = 2f(\alpha) \sum_{k=1}^{10} f(k)$$

$$= 2f(\alpha)(f(1)+f(2)+\dots+f(10)) \dots (2)$$

From (1)

$$f(2) = 2f(1) = 2^3$$

$$f(3) = 2f(2)f(1) = 2^5$$

$$\vdots$$

$$f(10) = 2^9 f(1) = 2^{19}$$

$$f(\alpha) = 2^{2\alpha-1}; \alpha \in \mathbb{N}$$

from (2)

$$\sum_{k=1}^{10} f(\alpha+k) = 2(2^{2\alpha-1})(2+2^3+2^5+\dots+2^{19})$$

$$\frac{512}{3}(2^{2\alpha}-1) = 2^{2\alpha} \left(2 \frac{(2^{20}-1)}{3} \right)$$

$$\text{Hence } \alpha = 4$$

Q4 (2)

$$f(x) = \left[2 \left(1 - \frac{x^{25}}{2} \right) (2 + x^{25}) \right]^{\frac{1}{50}}$$

$$f(x) = \left[(2 - x^{25})(2 + x^{25}) \right]^{\frac{1}{50}}$$

$$= (4 - x^{50})^{1/50}$$

$$f(f(x)) = \left(4 - \left((4 - x^{50})^{1/50} \right)^{50} \right)^{1/50} = x$$

$$g(x) = f(f(f(x))) + f(f(x))$$

$$= f(x) + x$$

$$g(1) = f(1) + 1 = 3^{1/50} + 1$$

$$[g(1)] = [3^{1/50} + 1] = 2$$

Q5 (B)

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$$f(x) = \frac{x-1}{x+1}$$

$$\Rightarrow f^2(x) = f(f(x)) = \frac{\frac{x-1}{x+1} - 1}{\frac{x-1}{x+1} + 1} = -\frac{1}{x}$$

$$f^3(x) = f(f^2(x)) = f\left(-\frac{1}{x}\right) = \frac{x+1}{1-x}$$

$$\Rightarrow f^4(x) = f\left(\frac{x+1}{1-x}\right) = -\frac{1}{x}$$

$$\Rightarrow f^6(x) = -\frac{1}{x} \Rightarrow f^6(6) = -\frac{1}{8}$$

$$f^7(x) = \left(-\frac{1}{x}\right) = \frac{x+1}{1-x}$$

$$\Rightarrow f^7(7) = \frac{8}{-6} = -\frac{4}{3}$$

$$\therefore -\frac{1}{6} + -\frac{4}{3} = -\frac{3}{2}$$

Q6 (D)

$$f(x) = x - 1; g(x) = \frac{x^2}{x^2 - 1}$$

$$f(g(x)) = g(x) - 1 = \frac{x^2}{x^2 - 1} - 1 = \frac{x^2 - x^2 + 1}{x^2 - 1}$$

$$f(g(x)) = \frac{1}{x^2 - 1}; x \neq \pm 1, \text{ even function}$$

→ Hence $f(g(x))$ is many one function

$$y = \frac{1}{x^2 - 1}$$

$$y \cdot x^2 - y = 1$$

$$x^2 = \left(\frac{1+y}{y}\right)$$

$$\left(\frac{1+y}{y}\right) \geq 0$$



Range:- $y \in (-\infty, -1] \cup (0, \infty)$

Hence, Range $\neq$ Co-domain $\Rightarrow f(g(x))$ is into

function

Q7 (99)

$$f(x) + f(1-x) = \frac{2e^{2x}}{e^{2x} + e} + \frac{2e^{2-2x}}{e^{2-ex} + e} = \left[\frac{e^{2x}}{e^{2x} + e} + \frac{e^2}{e^2 + e^{2x+1}} \right]$$

$$= 2 \left[\frac{e^{2x-1}}{e^{2x-1} + 1} + \frac{1}{1 + e^{2x-1}} \right] = 2$$

$$f\left(\frac{1}{100}\right) + f\left(\frac{2}{100}\right) + f\left(\frac{3}{100}\right) + \dots + f\left(\frac{99}{100}\right)$$

$$= \left\{ f\left(\frac{1}{100}\right) + f\left(\frac{99}{100}\right) \right\} + \left\{ f\left(\frac{2}{100}\right) + f\left(\frac{98}{100}\right) \right\} + \dots + \left\{ f\left(\frac{49}{100}\right) + f\left(\frac{51}{100}\right) \right\} + f\left(\frac{1}{2}\right)$$

$$= (2 + 2 + 2 + \dots - 49 \text{ times}) + \frac{2e}{e + e}$$

$$= 98 + 1 = 99$$

Q8 (190)

$$f^{-1}(n) = \begin{cases} \frac{n}{2} & ; n = 2, 4, 6, 8, 10 \\ \frac{n+11}{2} & ; n = 1, 3, 5, 7, 9 \end{cases}$$

$$f(g(n)) = \begin{cases} n+1 & ; n \in \text{odd} \\ n-1 & ; n \in \text{even} \end{cases}$$

$$\Rightarrow g(n) = \begin{cases} f^{-1}(n+1) & ; n \in \text{odd} \\ f^{-1}(n-1) & ; n \in \text{even} \end{cases}$$

$$\therefore g(n) = \begin{cases} \frac{n+1}{2} & ; n \in \text{odd} \\ \frac{n+10}{2} & ; n \in \text{even} \end{cases}$$

$$g(10) \cdot [g(1) + g(2) + g(3) + g(4) + g(5)]$$

$$= 10 \cdot [1 + 6 + 2 + 7 + 3] = 190$$

Q9 (D)

$$f(x) = \begin{cases} 4R & ; n = 2R \\ 4R - 2 & ; n = 4R - 1 \\ 2R - 1 & ; n = 4R - 3 \end{cases}$$

$(R \in \mathbb{N})$

Note that for any element, it will fall into exactly one of these sets.

$$\{y : y = 4R; y \in \mathbb{N}\}$$

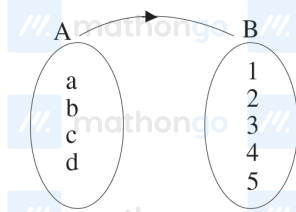
$$\{y : y = 4R - 2; y \in \mathbb{N}\}$$

$$\{y : y = 2R - 1; y \in \mathbb{N}\}$$

Corresponding to that y , we will get exactly one value of n .

Thus, f is one – one & onto.

Q10 (D)



$$n(s) = {}^5C_4 \times 4! = 120$$

f(a)	+	2f(b)	=	f(c)	+	f(d)
5		2×1		3		4
4		2×2		3		5
1		2×3		2		5

$$n(A) = 2 \times 3 = 6$$

$$\therefore P(A) = \frac{n(A)}{n(s)} = \frac{6}{120} = \frac{1}{20}$$

Q11 (37)

(1, 1), (1, 4), (4, 1), (2, 4), (4, 2), (3, 4), (4, 3),

(4, 4) – all have one choice for image.

(2, 1), (1, 2), (2, 2) – all have three choices for image

(3, 2), (2, 3), (3, 1), (1, 3), (3, 3) – all have two choices for image.

So the total functions = $3 \times 3 \times 2 \times 2 \times 2 = 72$

Case 1 : None of the pre-images have 3 as image

Total functions = $2 \times 2 \times 1 \times 1 \times 1 = 4$

Case 2 : None of the pre-images have 2 as image

Total functions = $2 \times 2 \times 2 \times 2 \times 2 = 32$

Case 3 : None of the pre-images have either 3 or 2 as image

Total functions = $1 \times 1 \times 1 \times 1 \times 1 = 1$

$\therefore$ Total onto functions = $72 - 4 - 32 + 1 = 37$

Q12 (D)

$$-1 \leq \frac{2 \sin^{-1} \left(\frac{1}{4x^2 - 1} \right)}{\pi} \leq 1$$

$$-\pi/2 \leq \sin^{-1} \frac{1}{4x^2 - 1} \leq \pi/2$$

Always $-1 \leq \frac{1}{4x^2 - 1} \leq 1$

$$x \in \left(\infty, \frac{1}{\sqrt{2}} \right) \cup \left[\frac{1}{\sqrt{2}}, \infty \right)$$

Q13 (3395)

$$f(x) = (c+1)x^2 + (1-c^2)x + 2k \quad \dots(1)$$

$$\& f(x+y) = f(x) + f(y) - xy \quad \forall xy \in \mathbb{R}$$

$$\lim_{y \rightarrow 0} \frac{f(x+y) - f(x)}{y} = \lim_{y \rightarrow 0} \frac{f(y) - xy}{y} \Rightarrow f'(x) = f'(0) - x$$

$$f(x) = -\frac{1}{2}x^2 + f'(0)x + \lambda \quad \text{but } f(0) = 0 \Rightarrow \lambda = 0$$

$$f(x) = -\frac{1}{2}x^2 + (1-c^2)x \quad \dots(2)$$

$$\text{as } f'(0) = 1 - c^2$$

Comparing equation (1) and (2)

$$\text{We obtain, } c = -\frac{3}{2}$$

$$f(x) = -\frac{1}{2}x^2 - \frac{5}{4}x$$

$$\text{Now } \left| 2 \sum_{x=1}^{20} f(x) \right| = \sum_{x=1}^{20} x^2 + \frac{5}{2} \sum_{x=1}^{20} x$$

$$= 2870 + 525$$

$$= 3395$$

Q14 (18)

$$f(g(x)) = 8x^2 - 2x$$

$$g(f(x)) = 4x^2 + 6x + 1$$

$$\text{So, } g(x) = 2x - 1 \quad g(2) = 3$$

$$\& f(x) = 2x^2 + 3x + 1$$

$$f(2) = 8 + 6 + 1 = 15$$

Ans. 18

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